

Low-complexity fiber nonlinear distortion mitigation for long-haul optical transmission based on transformer and triplets

Jianyu Meng,^{a,*} Ju Cai,^{b,*} Hongbo Zhang,^{a,b,*} Min Zhang,^b Qianwu Zhang,^c Yongzhi Zhang,^d and Yunhan Xiao^d

^aSichuan University, Information Section of the Logistics Support Department, Chengdu, China

^bChengdu University of Information Technology, College of Communication Engineering, Chengdu, China

^cShanghai University, Key Laboratory of Specialty Fiber Optics and Optical Access Networks, Shanghai, China

^dProduction and Application of Plastic Optical Fiber National and Local Engineering Laboratory, Chengdu, China

ABSTRACT. We propose and demonstrate an intrachannel fiber nonlinear impairments compensation scheme based on the transformer and simplified triplets. In comprehensive numerical experiments with a transmission distance of 1200 km for 30-GBaud dual-polarization 16-ary quadrature amplitude modulation signals, the Q factor gains of the proposed scheme are 2.48, 1.73, 0.98, and 0.31 dB compared to chromatic dispersion compensation, digital back propagation using 1-steps-per-span split-step Fourier method (1-StSp DBP), deep neural network (DNN), and 10-StSp DBP, respectively. The implementation complexity of the proposed scheme is reduced by 65% and 43% compared to 10-StSp DBP and DNN, respectively.

© 2024 Society of Photo-Optical Instrumentation Engineers (SPIE) [DOI: [10.1117/1.OE.63.3.038105](https://doi.org/10.1117/1.OE.63.3.038105)]

Keywords: fiber nonlinear impairments compensation; transformer; simplified triplets; dual-polarization 16-ary quadrature amplitude modulation

Paper 20230818G received Sep. 4, 2023; revised Jan. 26, 2024; accepted Feb. 24, 2024; published Mar. 19, 2024.

1 Introduction

Capacity of optical networks grows by 60% per year due to the rapid development of the Internet applications.¹ Based on the linear Shannon theory, the capacity of optical networks is proportional to the signal-to-noise ratio (SNR). This means that the capacity can be increased by simply increasing signal-launched power. However, with the increase of signal-launched power, the fiber nonlinear impairments (NLI) become stronger and stronger.² This leaves the NLI as one of the major limitations to increasing capacity of long-haul optical fiber transmission systems.^{3,4}

To address this problem, numerous NLI compensation (NLC) algorithms have been proposed, such as digital back-propagation (DBP),^{5–8} perturbative nonlinear compensation (PNC),^{9–13} Volterra series nonlinear equalizer (VSNE),^{14,15} phase-conjugate twin wave,^{16,17} and machine learning (ML).^{18–24} Among these schemes, DBP and VSNE require to perform multiple fast Fourier transformation (FFT) and inverse FFT (IFFT) operations, which incurs a relatively higher computational complexity. The NLC performance of PCTM will be at the expense of 50% spectrum efficiency. The PNC is widely considered as a powerful and promising NLC algorithm since it enables single-step and symbol-rate processing.⁹ The ML-based NLC algorithm received attention for its good NLC performance and the potential to further decrease computational complexity.

*Address all correspondence to Jianyu Meng, mengjy@scu.edu.cn; Ju Cai, caiju@cuit.edu.cn; Hongbo Zhang, zhanghb@cuit.edu.cn

Neural network, one of the popular choices for ML, can directly learn from the received data itself to figure out the NLI without prior knowledge of the transmission link parameters.²¹ Based on the PNC and deep neural network (DNN), a system-independent NLC scheme is proposed in Refs. 21 and 22. Instead of emulating nonlinear Schrödinger equation, this scheme directly estimates the NLI by equalizing triplets. As a feedforward neural network without memory, in order to contain all the features, the dimension of the input layer must be enlarged, which incurs a relatively higher computational complexity. For further decreasing the computational complexity, a system-independent NLC scheme based on the bidirectional long short-term memory (Bi-LSTM) is proposed in Ref. 24. As a sequential computation, the Bi-LSTM has a higher time complexity and poor real-time performance. Recently, the transformer performed the best in modeling sequential data, becoming a popular model for ML.^{25–29} Moreover, the transformer enables to capture global dependencies between feature and label by adopting self-attention, which has a higher parallelism and computational efficiency compared to recurrent neural network.³⁰ Based on these merits, it is powerful and promising to compensate the NLI for long-haul and large-capacity optical transmission system.

In this paper, we propose an NLC scheme based on the transformer and simplified triplets. The proposed scheme not only decreases computational complexity but also achieves better NLC performance. Moreover, the proposed scheme can work without prior knowledge of the link parameters. For sake of comparison, chromatic dispersion compensation (CDC), DBP, and DNN are also performed. Our study demonstrates that the proposed scheme enables to achieve better performance with lower computational complexity compared to the DNN and conventional compensation scheme.

2 Principle of Transformer-Based Nonlinear Impairments Compensation

2.1 Simplified Triplets

For single-mode fiber (SMF), the propagation of a dual-polarization symbol can be described as^{7,8}

$$\frac{\partial u_{x/y}(z, t)}{\partial z} + \left(\frac{\alpha}{2} + j \frac{\beta_2}{2} \frac{\partial^2}{\partial t^2} - \frac{\beta_3}{6} \frac{\partial^3}{\partial t^3} \right) u_{x/y}(z, t) = j\gamma(|u_{x/y}(z, t)|^2 + |u_{y/x}(z, t)|^2) u_{x/y}(z, t), \quad (1)$$

where α , β_2 , β_3 , and γ are, respectively, attenuation, group velocity dispersion, the slope of the group velocity dispersion, and fiber nonlinear coefficient; $u_{x/y}(z, t)$ is the optical field of x or y polarization. Based on the PNC theory, the intrachannel NLIs for a dual-polarization symbol are expressed as⁹

$$\begin{cases} \Delta u_{k,x} = P_0^{3/2} \left(\sum_{m \neq 0, n \neq 0} X_{k+n} X_{k+m} X_{(k+m+n)}^* C_{m,n} + \sum_{m \neq 0, n} Y_{k+n} X_{k+m} Y_{(k+m+n)}^* C_{m,n} \right), \\ \Delta u_{k,y} = P_0^{3/2} \left(\sum_{m \neq 0, n \neq 0} Y_{k+n} Y_{k+m} Y_{(k+m+n)}^* C_{m,n} + \sum_{m \neq 0, n} X_{k+n} Y_{k+m} X_{(k+m+n)}^* C_{m,n} \right), \end{cases} \quad (2)$$

where $\Delta u_{k,x}$ and $\Delta u_{k,y}$ are, respectively, the intrachannel NLI for the k 'th symbol of x and y polarizations; P_0 is the signal-launched power; X_{k+m} and Y_{k+m} are, respectively, the $(k+m)$ 'th transmitted symbols of x and y polarizations; m and n denote symbol indices; $C_{m,n}$ represents the nonlinear perturbation coefficient, which is expressed as

$$\begin{cases} C_{m,n} = j \frac{8}{9} \frac{\gamma \tau^2}{\sqrt{3}|\beta_2|} E_1 \left(-j \frac{mnT^2}{\beta_2 h} \right) & m \neq 0 \text{ and } n \neq 0, \\ C_{m,n} = j \frac{8}{9} \frac{\gamma \tau^2}{\sqrt{3}|\beta_2|} \frac{1}{2} E_1 \left(\frac{(m-n)^2 T^2 \tau^2}{3|\beta_2|^2 h^2} \right) & m \text{ or } n = 0, \end{cases} \quad (3)$$

where τ , T , h , and $E_1(\bullet)$ are, respectively, the pulse width, the symbol duration, the transmission distance, and the exponential integral function. Therefore, the triplets are defined as

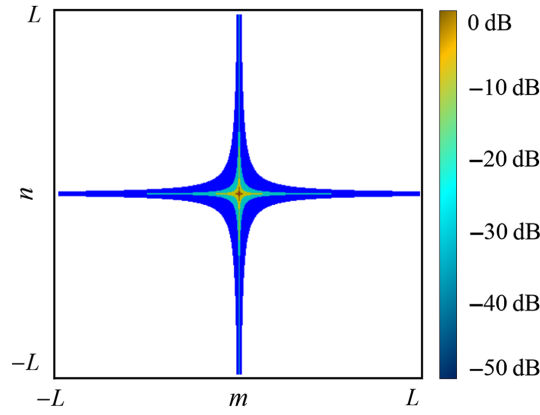


Fig. 1 Relative magnitude of $C_{m,n}$.

$$\begin{cases} T_{k,x} = X_{k+n}X_{k+m}X_{(k+m+n)}^* + Y_{k+n}X_{k+m}Y_{(k+m+n)}^*, \\ T_{k,y} = Y_{k+n}Y_{k+m}Y_{(k+m+n)}^* + X_{k+n}Y_{k+m}X_{(k+m+n)}^*, \end{cases} \quad (4)$$

where $T_{k,x}$ and $T_{k,y}$, are, respectively, the triplets for the k 'th symbol of x and y polarizations.

According to Eq. (4), the larger the value of m and n is, the greater the number of triplets, which incurs a higher computational complexity. To keep the complexity low, the number of triplets should be kept low. In conventional scheme, the $C_{m,n}$ were used to select triplets.²⁰ Figure 1 shows the relative magnitude of $C_{m,n}$ that is normalized to $C_{0,0}$, where the normalization is defined as $20 \lg(|C_{m,n}|/|C_{0,0}|)$. However, global ergodic of the conventional scheme led to relatively higher complexity. Thus this requires establishing a selection criterion that independent on the link parameters. As shown in Fig. 1, the $C_{m,n}$ has a hyperbolic dependence on the symbol indices, the simplified selection criterion is expressed as

$$\begin{cases} |mn| \leq G \\ -L \leq m, n \leq L \end{cases} \quad (5)$$

where L depends on memory length and G depends merely on implementation limit.²²

For further decreasing the computational complexity, we degenerate $X_{(k+m+n)}^*$ and $Y_{(k+m+n)}^*$ into QPSK symbols or the origin as shown in Fig. 2.¹² Therefore, complex multiplication between $X_{k+n}X_{k+m}$, $Y_{k+n}Y_{k+m}$, $X_{k+n}Y_{k+m}$, $Y_{k+n}X_{k+m}$, and $X_{(k+m+n)}^*$, $Y_{(k+m+n)}^*$ are replaced by a logical operation. Based on the above analysis, the intrachannel fiber nonlinear impairments for a dual-polarization symbol are expressed as^{9,12,13}

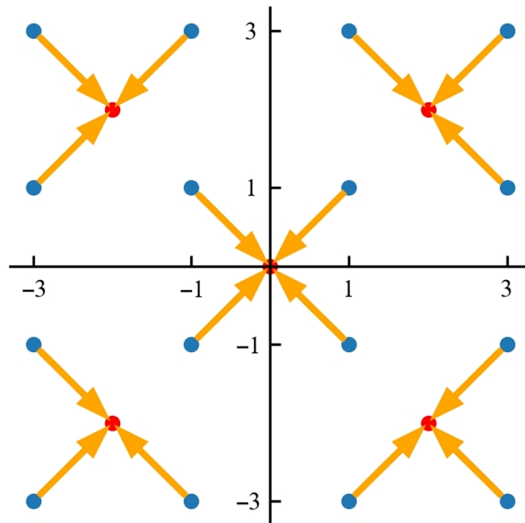


Fig. 2 Degenerated signal constellation for the 16-QAM symbol.

$$\begin{cases} \Delta u_{k,x} = P_0^{3/2} \sum_{q=1}^N \tilde{C}_q^N \left(\sum_{\substack{m \neq 0, n \neq 0 \\ C_{m,n} \approx \tilde{C}_q^N}} X_{k+n} X_{k+m} D(X_{(k+m+n)}^*) + \sum_{\substack{m \neq 0, n \\ C_{m,n} \approx \tilde{C}_q^N}} Y_{k+n} X_{k+m} D(Y_{(k+m+n)}^*) \right), \\ \Delta u_{k,y} = P_0^{3/2} \sum_{q=1}^N \tilde{C}_q^N \left(\sum_{\substack{m \neq 0, n \neq 0 \\ C_{m,n} \approx \tilde{C}_q^N}} Y_{k+n} Y_{k+m} D(Y_{(k+m+n)}^*) + \sum_{\substack{m \neq 0, n \\ C_{m,n} \approx \tilde{C}_q^N}} X_{k+n} Y_{k+m} D(X_{(k+m+n)}^*) \right), \end{cases} \quad (6)$$

where $\tilde{C}_q^N$ represents the quantized perturbation coefficients, N represents the quantization level, and $D(\bullet)$ represents the logical operation. According to Eqs. (4) and (6), the simplified triplets are expressed as

$$\begin{cases} T_{k,x} = X_{k+n} X_{k+m} D(X_{(k+m+n)}^*) + Y_{k+n} X_{k+m} D(Y_{(k+m+n)}^*), \\ T_{k,y} = Y_{k+n} Y_{k+m} D(Y_{(k+m+n)}^*) + X_{k+n} Y_{k+m} D(X_{(k+m+n)}^*). \end{cases} \quad (7)$$

2.2 Transformer

The transformer, introduced by Google Brain in 2017, replaces the recurrent neural network commonly used in NLP tasks by adopting self-attention mechanism and compensating for the missing positional information by adopting the positional encoding mechanism.³⁰ As shown in Fig. 3, the transformer model is composed of two parts: encoder and decoder. The encoder is used to encode the input sequence into a high-dimensional eigenvector, and the decoder is used to decode the high-dimensional vector into the target sequence. Among them, the encoder is composed of multihead attention layer and feed forward layer. The multihead attention layer maps query (Q), key (K), and value (V) through h different linear transformations to process data from different subspaces. Each attention is used for mapping features into different subspaces, which captures the dependencies between all features. The multihead attention mechanism can be expressed as³⁰

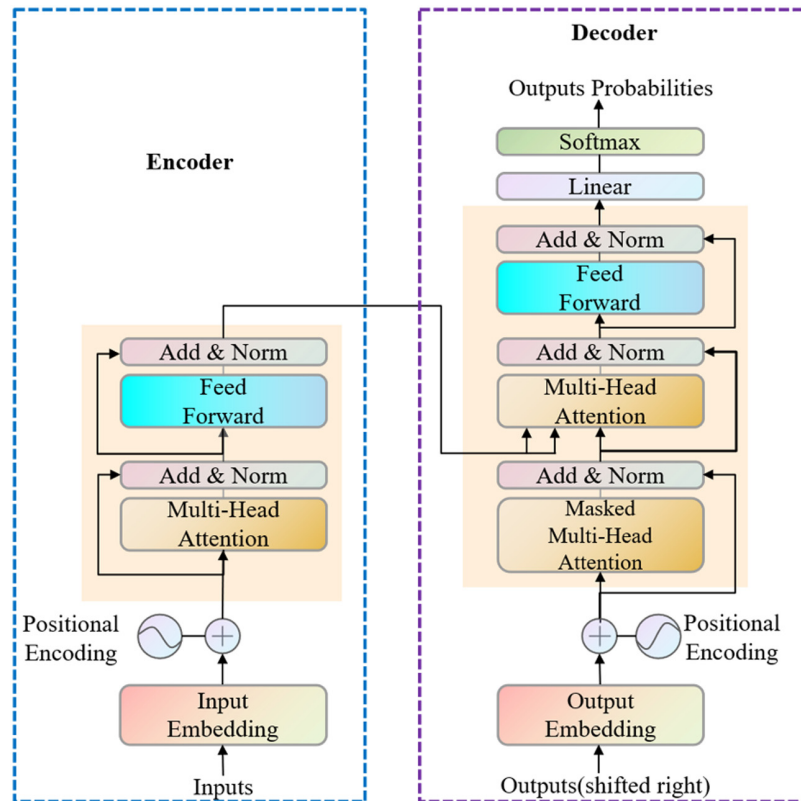


Fig. 3 Structure of the transformer.

$$\begin{cases} \text{multihead}(Q, K, V) = \text{concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W^O, \\ \text{head}_i = \text{attention}(QW_i^Q, KW_i^K, VW_i^V), \end{cases} \quad (8)$$

where $W_i^Q \in R^{d_{\text{model}} \times d_k}$, $W_i^K \in R^{d_{\text{model}} \times d_k}$, $W_i^V \in R^{d_{\text{model}} \times d_v}$, and $W_i^O \in R^{hd_v \times d_{\text{model}}}$ are the weight matrices, and $d_k = d_v = d_{\text{model}}/h$ and head_i denote i 'th attention. The feed forward layer, a simple fully connected network, is composed of two linear transformation layers and an RELU activation function, which can be expressed as

$$\text{FFN}(X) = \max(0, XW_1 + b_1)W_2 + b_2, \quad (9)$$

where X , W_i ($W_1 \in R^{d_{\text{ff}} \times d_{\text{ff}}}$ and $W_2 \in R^{d_{\text{ff}} \times d_{\text{model}}}$) and b_i are, respectively, the feature vector, the weight matrices, and the bias vector. Each sublayer of encoder is followed by an Add and Norm layer, where the Add layer is a residual connection and the Norm layer is a standard normalization layer.

It is worth noting that the encoder is key to the transformer model, and there is no additional benefit from the decoder.³¹ For further decreasing the implementation complexity of the proposed scheme, only the decoder is used here. The work of Ref. 32 shows that the self-attention can be replaced by cascaded linear layers and normalization layers. So the self-attention mechanism is canceled. Furthermore, the $C_{m,n}$ have taken into account the perturbation between symbols, so that there is no need for positional encoding for the transformer. Based on the above analysis, only the multihead attention, residual connection, and FFN are taken into account for the transformer.

During the training stage, the Adam optimizer, one of the popular choices for neural network, can rapid convergence compare with SGD.³³ However, we found that the Adam has higher specificity and poor generalization for different experimental setups. Recently, the work of Ref. 34 proposed AdaBound optimizer that trains as fast as Adam and as good as SGD, which is achieved by a bounded and smoothly transitioned between the required initial and final learning rates. Based on the above analysis, we set epoch to 200 and batch size to 256, and the optimizer used is AdaBound, mean square error is used for the loss function.

2.3 Nonlinear Impairments Compensation Based on Transformer and Simplified Triplets

The block diagram of coherent receiver DSP architecture with the proposed NLC scheme is shown in Fig. 4. First, the triplets are calculated according to Eq. (7), and then both the feature x_t and the label $y_{t,\text{label}}$ at the current time are input into the transformer. After the transformer network equalization, the estimated NLI is obtained, and the compensated symbol is obtained by subtracting from the original symbol. Finally, the bit error rate and Q factor of the transmission system are calculated, where Q factor = $20 \log_{10}(\sqrt{2}\text{erfc}^{-1}(2\text{BER}))$.

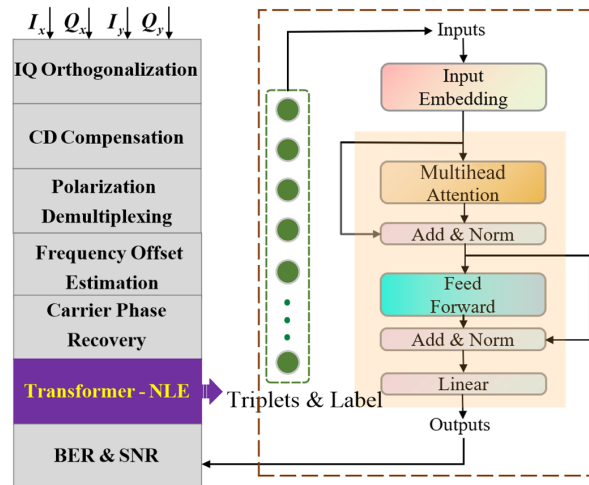


Fig. 4 Coherent receiver DSP architecture with the nonlinearity compensation based on transformer and simplified triplets.

2.4 Complexity Analysis

Since the computational complexity of real multiplication is 4 times that of addition operation, the number of real multiplications per transmitted symbol is considered as complexity of the NLC algorithm.²¹ Among them, the DBP algorithm is composed of CDC and NLC, where the computational complexity of the CDC algorithm can be expressed as

$$C_{\text{CDC}} = 4 \left(\frac{2M(\log_2(M) + 1)}{(M - M_{D_1} + 1)} + \frac{M(\log_2(M) + 1)}{(M - M_{D_2} + 1)} \right), \quad (10)$$

where M is the FFT size and $M_{D_q} = q\tau_D/T$, where τ_D is the dispersive channel impulse response, and q is the oversampling factor. Therefore, the computational complexity of DBP algorithm can be expressed as

$$C_{\text{DBP}} = 4N_{\text{sp}}N_{\text{StSp}} \left(\frac{M(\log_2(M) + 1)q}{(M - M_{D_q} + 1)} + q \right), \quad (11)$$

where N_{sp} and N_{StSp} are the total number of spans and the number of propagation steps per span, respectively. For DNN and transformer algorithm, the computational complexity is composed of triplets' computation and equalization processing. Among them, the computational complexity of DNN-NLC algorithm can be expressed as

$$C_{\text{DNN-NLC}} = 16N_{t,\text{DNN}} + \sum_{r=2}^R N_{r-1}N_r \quad (12)$$

where $N_{t,\text{DNN}}$, R , and N_r are the number of triplets of DNN, the total number of layers, and the number of neurons in the r 'th layer, respectively. The computational complexity of transformer-NLC algorithm using simplified triplets can be expressed as

$$C_{\text{transformer-NLC}} = 8N_{t,\text{transformer}} + d_i d_{\text{model}} + d_{\text{model}} d_o + N_{\text{layer}} (2d_{\text{model}}^2 + 2d_{\text{model}} d_{\text{ff}}), \quad (13)$$

where $N_{t,\text{transformer}}$, d_i , d_{model} , N_{layer} , and d_o are, respectively, the number of triplets of transformer network, the initial input dimension, the number of features, the number of the transformer layers, and the number of output units.

3 Results and Discussion

To demonstrate the excellent performance of the proposed NLC scheme, we construct that a 30-GBaud dual-polarization 16-ary quadrature amplitude modulation (DP-16QAM) coherent optical fiber transmission system is shown in Fig. 5. Table 1 presents the system parameters.

At the transmitter side, the bit sequences are generated by 15 repetitions of pseudorandom bit sequence (PRBS) with length 2^{16} . We use the first 32,768 symbols for training, the second 16,384 symbols for validation, and other 16,384 symbols for testing. Among them, the testing set contains a total of 245,760 (15 copies) symbols. First, the 30-GBaud DP-16QAM symbols with root rise cosine 0.01 are generated by 60 Gsample/s arbitrary waveform generator. Then the polarization beam splitter split 1550-nm optical carrier generated by external cavity laser into two polarized lights and then modulated by the corresponding IQ modulator to generate the

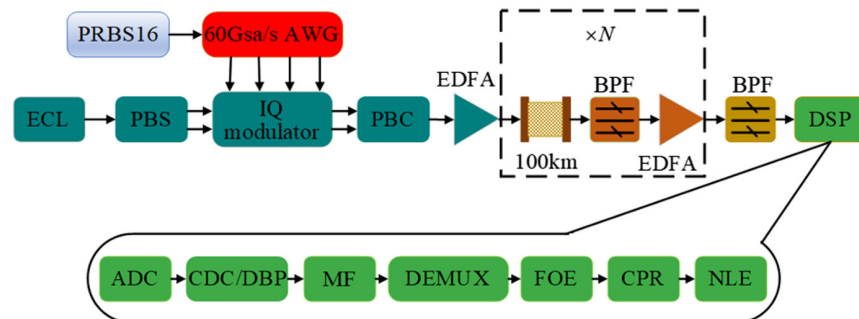


Fig. 5 Experimental setup for a single-channel transmission.

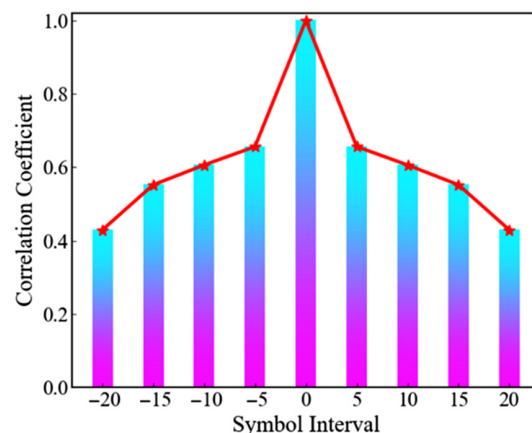
Table 1 Parameters of coherent optical fiber communication system.

Parameters	Value
Amplifier noise figure	6.0 dB/km
Attenuation	0.2 dB/km
Dispersion	16 ps/nm/km
Dispersion slope	0.08 ps/(nm ² · km)
Nonlinear coefficient	1.3 W ⁻¹ km ⁻¹

modulated signal. Finally, the polarization beam coupler combines two modulated signals into an optical signal and then the signal is power amplified by the erbium-doped optical fiber amplifier (EDFA) and launch into the fiber link. At the fiber link side, the link is implemented via fiber loops, which consists of 100 km SMF, EDFA, band-pass filter (BPF), and loop controller. A flat-top optical filter is used in each span to mitigate the EDFA spontaneous emission noise. At the receiver side, a 90-deg optical hybrid, four 43-GHz balanced photodiodes, and an 80-GSa/s real-time sampling oscilloscope were used to receive the signal. The received signal is first filters out the out-of-band noise via BPF, and then CDC or DBP, match filter, polarization demultiplexing with modified cascaded multimodulus algorithm,³⁵ frequency offset estimation (FOE) with FFT-FOE algorithm,³⁶ carrier phase recovery (CPR) with modified blind phase search algorithm³⁷ and the proposed NLC scheme is performed.

After the CPR process, the triplets are calculated according to Eq. (7). In the actual transmission process, symbols can overlap and interact with thousands of adjacent symbols. Therefore, considering only the NLI of the current symbol will worsen the performance of the transmission system. Figure 6 shows the correlation coefficients of the symbols after CPR process. It can be seen that the current symbol has the largest correlation, and with the time interval increases, the correlation decreases. The interaction between linear and nonlinear impairments can be further mitigated by selecting an appropriate L . If the L is too large, redundancy will be introduced and the processing speed will be decreased, and the performance will become worse if too small. Based on Fig. 6, L take 15 for the generating the features required by transformer, and the performance and computational complexity are balanced by adjusting the G .

The Q factor versus G for 30-GBuad DP-16-QAM after 1200-km transmission is evaluated as shown in Fig. 7. As shown in Fig. 7, with the value of G increases, the NLC performance tends to stabilize. Compared to $G = 13$ and $G = 15$, $G = 11$ can separately improves the Q factor by 0.11 and 0.08 dB with has a smaller memory length. Compared to $G = 9$, although the $G = 11$ has a larger memory length, improves the Q factor by 0.11 dB and the optimal signal-launched power by 1 dBm. Considering the performance and the memory length, we select $L = 15$ and $G = 11$ for generating the features required by the transformer network.

**Fig. 6** Signal sequence correlation.

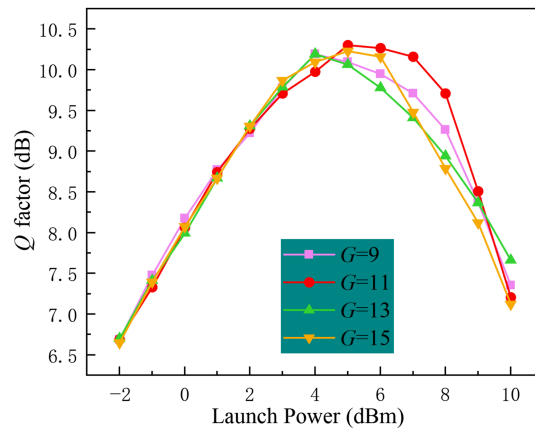


Fig. 7 Q factor versus G for 30-GBaud DP-16QAM signals after 1200-km transmission.

Figure 8 plots the comparison of Q factor as a function of signal-launched power for the proposed NLC scheme after 1200-km transmission. For demonstrating the excellent performance of the proposed NLC scheme, we also perform CDC, DBP using 1 steps-per-span split-step Fourier method (1-StSp DBP), 10-StSp DBP, and DNN scheme. The number of neurons per layer in DNN is 2058, 2, 10, and 2. As shown in Fig. 8, the proposed NLC scheme has the largest Q factor at 5 dBm signal-launched power, reaching 10.31 dB. Compared to CDC scheme, 1-StSp DBP, DNN, 10-StSp DBP, and transformer not only increase the optimal signal-launched power, also the Q factor improve by 0.75, 1.50, 2.17, and 2.48 dB, respectively. It can be seen that the Q factor gains of the proposed scheme is 1.73 and 0.31 dB compared to 1-StPS DBP and 10-StPS DBP, respectively. Furthermore, the proposed NLC scheme can work without prior knowledge of the link parameters.

For further demonstrating the excellent performance of the proposed NLC scheme, the SNR dependence on transmission distance at the optimal signal-launched power is shown in Fig. 9. It can be seen that the SNR of these NLC scheme will deteriorate with the increase of transmission distance. This is mainly because EDFA is required to be used for relay amplification after each span transmission, which introduce EDFA spontaneous emission noise. Due to the NLC performance of the proposed NLC scheme is more excellent, the SNR declines particularly slowly.

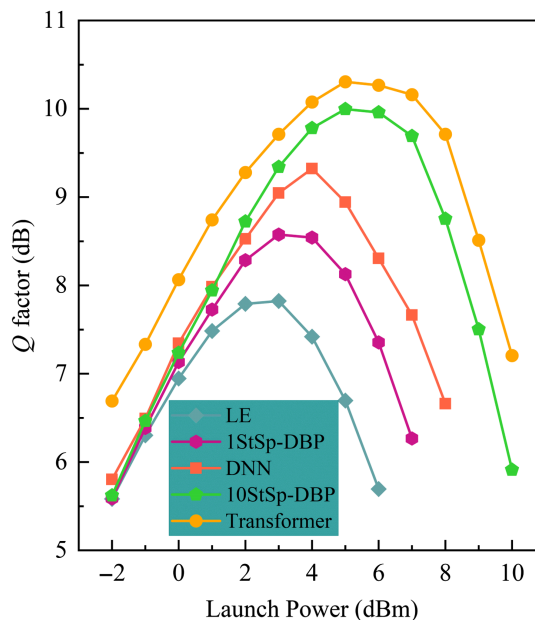


Fig. 8 Q factor versus signal-launched power for the proposed NLC scheme after 1200-km transmission.

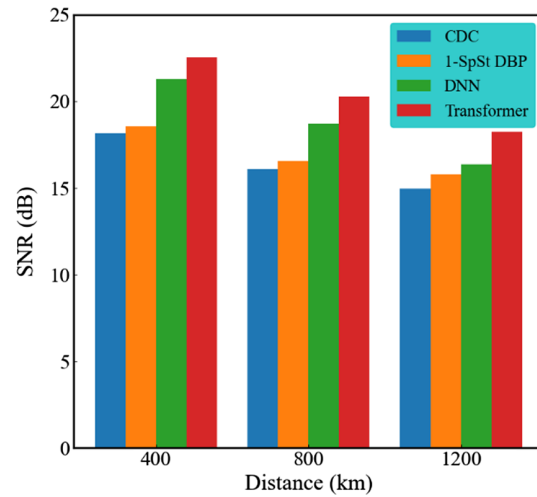


Fig. 9 SNR versus transmission distance for the proposed NLC scheme at the optimal signal-launched power.

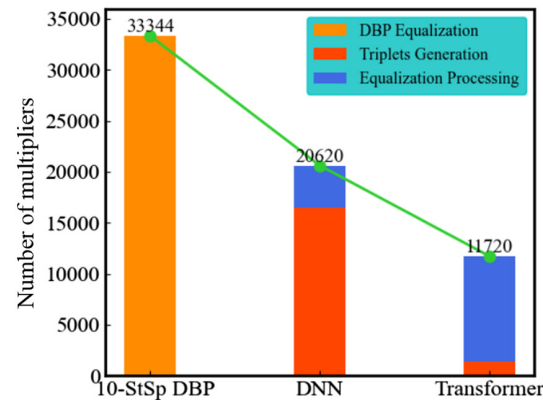


Fig. 10 Computational complexity comparison of NLC algorithms.

It proves that the proposed NLC scheme is more suitable for the application in the long-haul optical fiber transmission system.

Based on the transmission link parameters and the detailed neural network parameters, the computational complexity comparison of NLC scheme based on DNN, transformer, and 10-StSp DBP is summarized in Fig. 10. Significantly, such comparisons do not take into account the complexity of training, which is usually performed offline. As shown in Fig. 10, the number of multipliers required by transformer-NLC is about 35% and 57% of 10-StSp DBP and DNN, respectively. Considering the performance and complexity of NLC scheme, the proposed NLC scheme has a better application prospect for long-haul optical transmission system.

4 Conclusion

In this paper, we propose and demonstrate a NLC scheme based on transformer and simplified triplets. The experiment results indicate that the Q factor gains of the proposed scheme is 2.48, 1.73, 0.98, and 0.31 dB compared to CDC, 1-StSp DBP, DNN, and 10-StSp DBP, respectively. The implementation complexity of the proposed scheme is reduced by 65% and 43% compared to 10-StSp DBP and DNN, respectively. At the training stage for neural network, it is necessary to have sufficient nonlinearity. Combining with some other features, the proposed scheme is expected to further maintain the advantage of low complexity. Such study is left for future research. In addition, the proposed scheme independent of the transmission link, and can works without prior knowledge of the link parameters.

Code and Data Availability

Our DSP is publicly available at <https://github.com/JackMengJian/DSP>. For the raw NLE, please reach out to the authors.

Acknowledgments

This work is supported by the Sichuan Science and Technology Program (Grant No. 2021YFG0149), and by the Science and Technology Commission of Shanghai Municipality (Grant No. SKLSFO2019-06).

References

1. P. J. Winzer, D. T. Neilson, and A. R. Chraplyvy, "Fiber-optic transmission and networking: the previous 20 and the next 20 years [Invited]," *Opt. Express* **26**(18), 24190–24239 (2018).
2. A. D. Ellis, J. Zhao, and D. Cotter, "Approaching the non-linear Shannon limit," *J. Lightwave Technol.* **28**(4), 423–433 (2010).
3. K. Roberts et al., "Nonlinear mitigation enabling next generation high speed optical transport beyond 100G," in *Opt. Fiber Commun. Conf. (OFC)*, p. M3J.1 (2019).
4. Q. Zhuge et al., "Fiber nonlinearity mitigation and compensation for capacity-approaching optical transmission systems," in *Asia Commun. and Photonics Conf. (ACP)*, pp. 1–3 (2019).
5. E. Ip and J. M. Kahn, "Compensation of dispersion and nonlinear impairments using digital backpropagation," *J. Lightwave Technol.* **26**(20), 3416–3425 (2008).
6. J. Gonçalves et al., "Nonlinear compensation with DBP aided by a memory polynomial," *Opt. Express* **24**(26), 30309–30316 (2016).
7. S. Rahbarfam and S. Kumar, "Nonlinear phase noise reduction using digital back propagation and midpoint optical phase conjugation," *Opt. Express* **27**(6), 8968–8982 (2019).
8. N. V. Dien et al., "Digital back-propagation optimization for high-baudrate single-channel optical fiber transmissions," *Opt. Commun.* **491**, 126913 (2021).
9. Z. Tao et al., "Multiplier-free intrachannel nonlinearity compensating algorithm operating at symbol rate," *J. Lightwave Technol.* **29**(17), 2570–2576 (2011).
10. L. Dou et al., "Pre-distortion method for intra-channel nonlinearity compensation with phase-rotated perturbation term," in *OFC/NFOEC*, pp. 1–3 (2012).
11. Q. Zhuge et al., "Aggressive quantization on perturbation coefficients for nonlinear pre-distortion," in *OFC 2014*, pp. 1–3 (2014).
12. T. Oyama et al., "Nonlinear compensation technologies for future optical communication systems," *Proc. SPIE* **9389**, 93890I (2015).
13. Z. Li et al., "MMSE-based optimization of perturbation coefficients quantization for fiber nonlinearity mitigation," *J. Lightwave Technol.* **33**(20), 4311–4317 (2015).
14. F. P. Guiomar et al., "Experimental demonstration of a frequency-domain Volterra series nonlinear equalizer in polarization-multiplexed transmission," *Opt. Express* **21**(1), 276–288 (2013).
15. G. Sharan Yadav et al., "Reducing computation complexity by using elastic net regularization based pruned Volterra equalization in a 80 Gbps PAM-4 signal for inter-data center interconnects," *Opt. Express* **28**(26), 38539–38552 (2020).
16. X. Liu et al., "Phase-conjugated twin waves for communication beyond the Kerr nonlinearity limit," *Nat. Photonics* **7**(7), 560–568 (2013).
17. S. S. Husain and J. K. Hmood, "Performance enhancement of PCTWs technique by employing RZ-coding in phase-modulated optical communication systems," *AIP Conf. Proc.* **2213**(1), 020094 (2020).
18. J. Zhang et al., "Non-data-aided k-nearest neighbors technique for optical fiber nonlinearity mitigation," *J. Lightwave Technol.* **36**(17), 3564–3572 (2018).
19. O. Sidelnikov, A. Redyuk, and S. Sygletos, "Equalization performance and complexity analysis of dynamic deep neural networks in long haul transmission systems," *Opt. Express* **26**(25), 32765–32776 (2018).
20. V. Kamalov et al., "Evolution from 8-QAM live traffic to PS 64-QAM with neural-network based nonlinearity compensation on 11000 km open subsea cable," in *Opt. Fiber Commun. Conf. Postdeadline Pap.*, p. Th4D.5 (2018).
21. S. Zhang et al., "Field and lab experimental demonstration of nonlinear impairment compensation using neural networks," *Nat. Commun.* **10**(1), 3033 (2019).
22. Y. Gao et al., "Reduced complexity nonlinearity compensation via principal component analysis and deep neural networks," in *Opt. Fiber Commun. Conf. (OFC)*, p. Th2A.49 (2019).
23. X. Liu, Y. Wang, and C. Li, "Nonlinear equalizer by feature engineering based-deep neural network for coherent optical communication system," in *Asia Commun. and Photonics Conf. (ACP) and Int. Conf. Inf. Photonics and Opt. Commun. (IPOC)*, pp. 1–3 (2020).

24. H. Yang et al., "Fast and accurate waveform modeling of long-haul multi-channel optical fiber transmission using a hybrid model-data driven scheme," *J. Lightwave Technol.* **40**(14), 4571–4580 (2022).
25. K. Huang et al., "MolTrans: molecular interaction transformer for drug–target interaction prediction," *Bioinformatics* **37**(6), 830–836 (2021).
26. H. Wu et al., "Autoformer: decomposition transformers with auto-correlation for long-term series forecasting," arXiv:2106.13008 (2022).
27. J. Chu, J. Cao, and Y. Chen, "An ensemble deep learning model based on transformers for long sequence time-series forecasting," in *Neural Comput. for Adv. Appl.*, pp. 273–286 (2022).
28. B. Li et al., "DRAformer: differentially reconstructed attention transformer for time-series forecasting," arXiv:2206.05495 (2022).
29. Y. Takagi et al., "Transformer-based personalized attention mechanism for medical images with clinical records," *J. Pathol. Inf.* **14**, 100185 (2023).
30. A. Vaswani et al., "Attention is all you need," arXiv:1706.03762 (2023).
31. T. Domhan, "How much attention do you need? A granular analysis of neural machine translation architectures," in *Proc. 56th Annu. Meet. of the Assoc. for Comput. Linguistics (Volume 1: Long Papers)* (2018).
32. M.-H. Guo et al., "Beyond self-attention: external attention using two linear layers for visual tasks," *IEEE Trans. Pattern Anal. Mach. Intell.* **45**(5), 5436–5447 (2023).
33. D. P. Kingma and J. Ba, "Adam: a method for stochastic optimization," arXiv:1412.6980 (2017).
34. L. Luo et al., "Adaptive gradient methods with dynamic bound of learning rate," arXiv:1902.09843 (2019).
35. M. Zhang et al., "A novel scalar MCMMA blind equalization utilized in 8-PAM LED based visible light communication system," in *IEEE Int. Conf. Commun. Workshops (ICC)*, pp. 321–325 (2016).
36. Y. Wang, K. Shi, and E. Serpedin, "Non-data-aided feedforward carrier frequency offset estimators for QAM constellations: a nonlinear least-squares approach," *EURASIP J. Adv. Signal Process.* **2004**(13), 1–9 (2004).
37. Y. Chen and X. Huang, "A simple three-stage carrier phase estimation algorithm for 16-QAM systems based on QPSK partitioning and maximum likelihood detection," *Opt. Commun.* **342**, 44–50 (2015).

Jianyu Meng is an engineer at Sichuan University, Chengdu, China. His current research interests include optical performance monitoring, high-speed optical communications systems, and nonlinear compensation algorithm of coherent optical communication system based on neural network.

Ju Cai is a professor at the Communication Engineering College, Chengdu University of Information Technology, Chengdu, China. His main research interests include high-speed optical communications systems, optical access networks, and nonlinear compensation algorithm of coherent optical communication system based on neural network.

Hongbo Zhang is an associate professor at the Communication Engineering College, Chengdu University of Information Technology, Chengdu, China. He received his PhD from University of Electronic Science and Technology of China. His current research interests include high-speed optical communications systems, modern communication networks, and nonlinear compensation algorithm of coherent optical communication system based on neural network.

Biographies of the other authors are not available.